

Step 1 Results Note

Baseline OCR Performance Across Synthetic Degradation Levels

Field	Detail
Problem ID	ML-T2-022
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Stage	Stage 2, Step 1 — Baseline OCR (faded-text track, no restoration)

1. Purpose

Establish a plain OCR baseline (no restoration or reconstruction) on synthetically degraded text, per the Stage 2 build philosophy of validating each pipeline component independently before integration. This baseline is the reference point every later approach (B, C, D) must be measured against.

2. Method

- Ground truth text rendered as a clean image (black text, white background, Arial 24pt), sized exactly to the text using font metrics rather than a fixed guessed canvas.
- Three synthetic degradation presets applied (mild, moderate, severe), each combining contrast reduction, Gaussian blur, and pixel-level random noise at increasing intensity.
- 5 independently-seeded samples generated per severity level (15 images total), each seed fixed and saved for reproducibility.
- OCR run via Tesseract (--psm 6, single uniform text block) on every image.
- CER and WER calculated per image using jiwer, against the known ground truth text.

3. Results

Level	CER (mean / min / max)	WER (mean / min / max)
Mild	0.000 / 0.000 / 0.000	0.000 / 0.000 / 0.000
Moderate	1.702 / 1.154 / 2.854	3.421 / 2.105 / 5.789
Severe	0.998 / 0.731 / 1.508	1.653 / 1.000 / 3.105

Mild degradation was read perfectly by Tesseract across all 5 seeds (CER/WER = 0.000). Moderate degradation produced a higher mean error than severe degradation, and scored worse on 4 of 5 seed-to-seed comparisons; two moderate samples

(seed 2 and seed 3) produced OCR output roughly 2.5–3.5x longer than the ground truth text, indicating hallucinated content rather than simple misreading.

4. Qualitative Check

The two flagged moderate images (seed 2, seed 3) were visually inspected against other moderate samples that scored substantially better. No meaningful visual difference was found, all moderate-level images remain clearly human-readable and represent a plausible faded/poor scan. The degradation preset itself is not the cause of the anomaly; the specific noise pattern in those seeds appears to interact with Tesseract's layout/character detection in a way that triggers runaway false-positive text generation.

5. Interpretation

This is treated as a diagnostic observation at this stage, not a fully established finding, given the small sample size (5 seeds per level, single sentence of text). With that caveat, the result is consistent and repeated rather than a single anecdote: moderate degradation produced worse mean CER/WER than severe degradation, and the failure mode differed qualitatively between levels, severe degradation mostly caused Tesseract to fail silently (little or no output), while moderate degradation more often caused Tesseract to hallucinate substantial false content while appearing, to a human, no less readable than other moderate samples that scored well.

This directly motivates the uncertainty-tagging component (Approach C) planned for later in Stage 2: a fixed, human-readable degradation level can produce highly unstable and unpredictable OCR failure behavior, including confident-looking but entirely fabricated output, which is precisely the risk that confidence scoring and uncertainty flagging are intended to catch rather than silently pass through as fact.

6. Next Steps

Baseline OCR (Step 1) is considered complete for this stage. Proceeding to Step 2: synthetic torn/missing-region mask generation, per the Stage 2 development plan.